Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.4 – Parallel Circuits

Unit 2: Electricity & Energy

The question: *last lesson, every component shared the one path round the loop. What changes if the current gets a choice?* A **junction** is a point where a wire splits into two. Today's circuits all have one: that single fact decides both rules below, and both rules are the **opposite** of series.

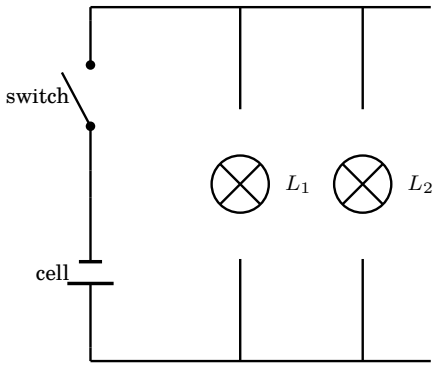
Learning intentions

All students will state and apply the **current rule** and the **voltage rule** for parallel circuits.
Most students will explain why parallel bulbs stay at full brightness however many branches you add.
Some students will calculate a missing current using both rules together.

Part 1 — The current rule

At a junction, the current **splits** between the branches. None of it is lost, so:

The branch currents ADD UP to the total current leaving the cell.



1) **Draw** an ammeter in the branch with L_1, one in the branch with L_2, and one more on the left leg, below the switch, to read the **total** current.

An ammeter symbol breaks the L_1 rung in series with the bulb; another breaks the L_2 rung in series with its bulb; a third breaks the left leg between the switch and the bottom bar, where all the current (not yet split) still flows.

2) **Predict** the current through each bulb, if the total current is **0.6 A** and L_1 and L_2 are identical bulbs.

Model answer: *0.3 A through each — the total splits evenly only because the two bulbs are identical and offer the current an equal choice.*

Build it

Build the circuit and measure the current through L_1, through L_2, and the total current leaving the cell.

$L_1 =$ _____ A $L_2 =$ _____ A total = _____ A

Check: $L_1 + L_2$ should add up close to the total.

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Part 2 — The voltage rule

Both branches connect to the **same** two points — the top bar and the bottom bar. Since voltage is about the difference between two points, and both branches span the *same* two points:

Every branch has the SAME voltage across it — equal to the cell's voltage.

3) **State** the voltage across L_2, if the cell is **9 V** and the voltage across L_1 is **9 V**.

Model answer: *9 V — the same as the cell and the same as L_1. Every branch in a parallel circuit sees the full cell voltage.*

Build it

Measure the voltage across the cell, across L_1, and across L_2.

Cell = _____ V L_1 = _____ V L_2 = _____ V

Check: all three readings should be close to equal.

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Most students — adding a third branch

4) **Explain** what happens to the brightness of L_1 and L_2 if you add a **third** bulb as a new branch, in parallel with the other two. Use the voltage rule in your answer — and say what *does* change, even though the brightness does not.

Model answer: *L_1 and L_2 stay exactly as bright as before — the voltage rule says every branch still gets the full cell voltage, with or without a third branch, so neither bulb notices it is there. What changes is the **total current** the cell has to supply — it goes up, because now three branches are drawing current instead of two.*

Some students — using both rules together

5) **Calculate** the current through L_2, if the total current is **0.5 A** and the current through L_1 is **0.35 A**.

Model answer: $0.5 - 0.35 = 0.15 A$ *through* L_2.

Part 3 — On your own

The sockets and lights in a house are wired in **parallel**, not series.

6) **Explain** why this matters in two ways: what happens in the rest of the house if one lamp blows, and what happens to the brightness of the other lamps if you plug in one more appliance. Use the current rule *and* the voltage rule.

Model answer: *Voltage rule: every socket and light is its own branch, so each still gets the full supply voltage even if one lamp blows — the rest of the house is unaffected, unlike a series string where one break kills everything. Current rule: plugging in one more appliance adds a new branch, so the **total** current from the supply goes up, but each existing branch still has the same voltage across it as before — so the other lamps stay exactly as bright.*

HOMEWORK

Make a two-column table, **Series** and **Parallel**. Write one sentence in each column for the current rule, and one sentence in each column for the voltage rule — four sentences in total.